Exercício 4.2: Distribuição garantida com anti-affinity e topologySpreadConstraints

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Ambiente

- Cluster kind v0.32.0 com 3 nós, Kubernetes v1.36.1, Calico v3.33.0, namespace `store-dev`. O enunciado pede `minikube start --nodes=3`; o kind com 3 nós é o equivalente usado em todo o trabalho.
- No minikube o control-plane é agendável. No kind ele vem com taint `NoSchedule`, então retirei o taint para contar como a terceira "zona", como sugere a dica.
- Cada cenário apaga o Deployment anterior antes de aplicar o próximo (mesmo nome `store-api`). Sequência exata em `04-commands.sh`.
- Nas saídas de `get pods -o wide` as colunas vazias `NOMINATED NODE` e `READINESS GATES` foram omitidas para caber na página.

Rótulos de zona

```
$ kubectl taint node fiap-store-control-plane node-role.kubernetes.io/control-plane:NoSchedule-
node/fiap-store-control-plane untainted
$ kubectl label node fiap-store-control-plane topology.kubernetes.io/zone=za
$ kubectl label node fiap-store-worker topology.kubernetes.io/zone=zb
$ kubectl label node fiap-store-worker2 topology.kubernetes.io/zone=zc

$ kubectl get nodes -L topology.kubernetes.io/zone
NAME                       STATUS   ROLES           AGE   VERSION   ZONE
fiap-store-control-plane   Ready    control-plane   10m   v1.36.1   za
fiap-store-worker          Ready    <none>          10m   v1.36.1   zb
fiap-store-worker2         Ready    <none>          10m   v1.36.1   zc
```

Tabela comparativa

Configuração	6 réplicas (cp / worker / worker2)	Pending em 6	10 réplicas (cp / worker / worker2)	Pending em 10
(A) anti-affinity `preferred` por hostname	2 / 2 / 2	0	3 / 4 / 3	0
(B) topologySpread zona `DoNotSchedule` + hostname `ScheduleAnyway`	2 / 2 / 2	0	3 / 4 / 3	0
(C) anti-affinity `required` por hostname	1 / 1 / 1	3	1 / 1 / 1	7

- **(A)** `preferred` é só peso na pontuação do scheduler: espalha enquanto dá e, quando todo nó já tem um Pod, empilha sem bloquear. Nada fica Pending, mas também nada garante o equilíbrio.
- **(B)** `maxSkew: 1` por zona com `DoNotSchedule` é regra dura: a diferença entre a zona mais cheia e a mais vazia nunca passa de 1. Em 6 fica exatamente 2/2/2 e em 10 fica 3/4/3 (skew 1), sem Pending.
- **(C)** `required` por hostname permite no máximo 1 Pod por nó. Com 3 nós, só 3 réplicas sobem e o resto fica Pending para sempre. Já em 6 réplicas sobram 3 Pending; em 10, 7.

Manifestos

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: store-api
  namespace: store-dev
  labels:
    app: store-api
spec:
  replicas: 6
  selector:
    matchLabels:
      app: store-api
  template:
    metadata:
      labels:
```

```
        app: store-api
    spec:
      affinity:
        podAntiAffinity:
          preferredDuringSchedulingIgnoredDuringExecution:
            - weight: 100
              podAffinityTerm:
                topologyKey: kubernetes.io/hostname
                labelSelector:
                  matchLabels:
                    app: store-api
      containers:
        - name: store-api
          image: store-api:1.0
          imagePullPolicy: IfNotPresent
          ports:
            - name: http
              containerPort: 8000
          readinessProbe:
            httpGet:
              path: /ready
              port: http
          resources:
            requests:
              cpu: 50m
              memory: 64Mi
```

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: store-api
  namespace: store-dev
  labels:
    app: store-api
spec:
  replicas: 6
  selector:
    matchLabels:
      app: store-api
  template:
    metadata:
      labels:
        app: store-api
    spec:
      topologySpreadConstraints:
        - maxSkew: 1
          topologyKey: topology.kubernetes.io/zone
```

```
      whenUnsatisfiable: DoNotSchedule
      labelSelector:
        matchLabels:
          app: store-api
    - maxSkew: 1
      topologyKey: kubernetes.io/hostname
      whenUnsatisfiable: ScheduleAnyway
      labelSelector:
        matchLabels:
          app: store-api
  containers:
    - name: store-api
      image: store-api:1.0
      imagePullPolicy: IfNotPresent
      ports:
        - name: http
          containerPort: 8000
      readinessProbe:
        httpGet:
          path: /ready
          port: http
      resources:
        requests:
          cpu: 50m
          memory: 64Mi
```

Em `03-required.yaml` muda só o bloco de afinidade:

```
  affinity:
    podAntiAffinity:
      requiredDuringSchedulingIgnoredDuringExecution:
        - topologyKey: kubernetes.io/hostname
          labelSelector:
            matchLabels:
              app: store-api
```

Cenário (A): podAntiAffinity preferred

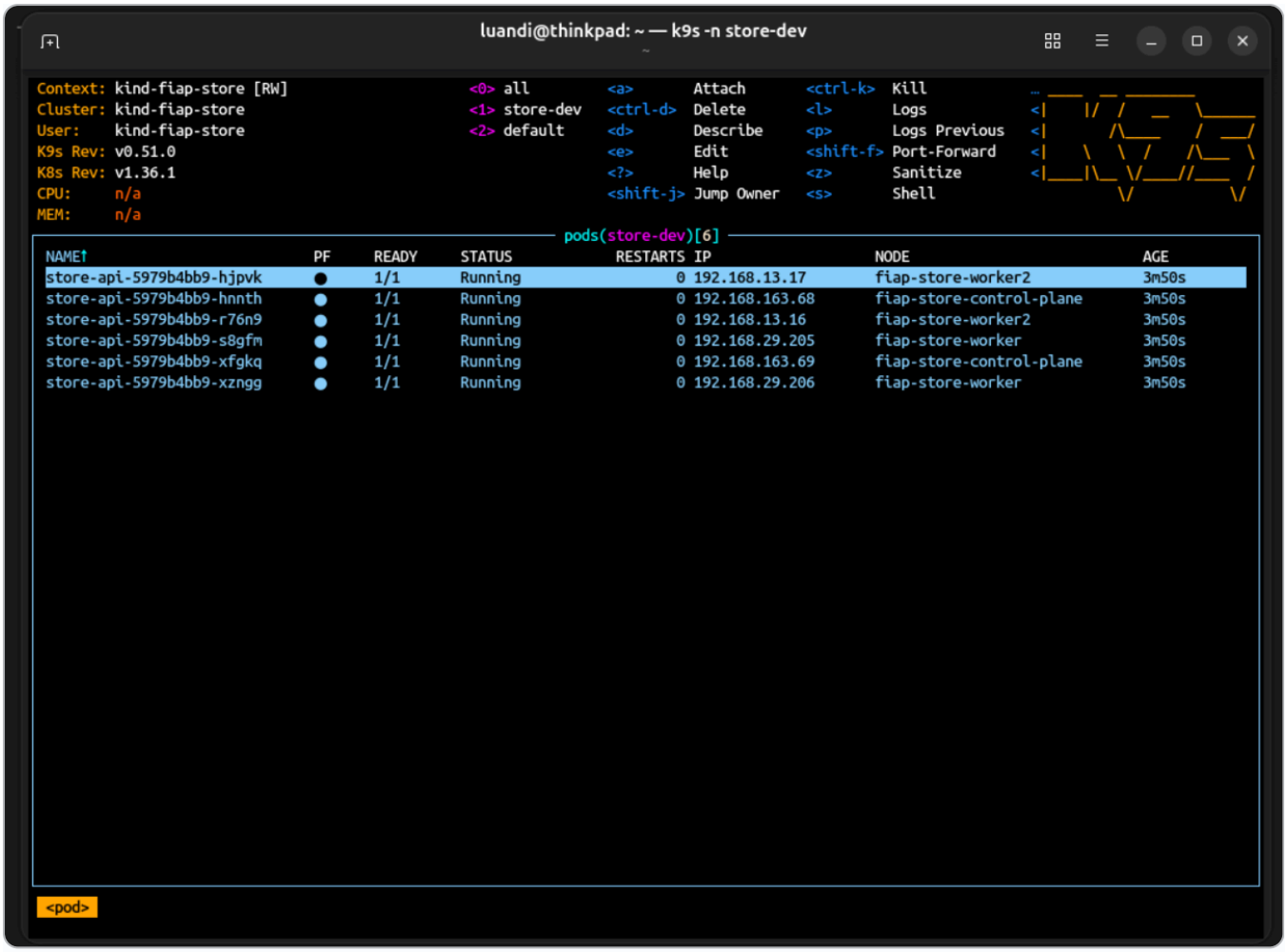
```
$ kubectl -n store-dev get pods -o wide --sort-by=.spec.nodeName   # 01-antiaffinity.yaml (6
replicas)
NAME                         READY   STATUS    RESTARTS   AGE   IP               NODE
store-api-55b44fb4b5-dqlx7   1/1     Running   0          13s   192.168.163.65   fiap-store-control-
plane
store-api-55b44fb4b5-nfxsq   1/1     Running   0          13s   192.168.163.66   fiap-store-control-
plane
store-api-55b44fb4b5-c7dfv   1/1     Running   0          13s   192.168.29.202   fiap-store-worker
store-api-55b44fb4b5-f9b2s   1/1     Running   0          13s   192.168.29.201   fiap-store-worker
store-api-55b44fb4b5-55sr6   1/1     Running   0          13s   192.168.13.13    fiap-store-worker2
store-api-55b44fb4b5-m767m   1/1     Running   0          13s   192.168.13.14    fiap-store-worker2

$ kubectl -n store-dev get pods -o wide --sort-by=.spec.nodeName   # 01-antiaffinity.yaml (10
replicas)
NAME                         READY   STATUS    RESTARTS   AGE   IP               NODE
store-api-55b44fb4b5-dqlx7   1/1     Running   0          26s   192.168.163.65   fiap-store-control-
plane
store-api-55b44fb4b5-h28dr   1/1     Running   0          13s   192.168.163.67   fiap-store-control-
plane
store-api-55b44fb4b5-nfxsq   1/1     Running   0          26s   192.168.163.66   fiap-store-control-
plane
store-api-55b44fb4b5-c7dfv   1/1     Running   0          26s   192.168.29.202   fiap-store-worker
store-api-55b44fb4b5-f9b2s   1/1     Running   0          26s   192.168.29.201   fiap-store-worker
store-api-55b44fb4b5-rnxxw   1/1     Running   0          13s   192.168.29.204   fiap-store-worker
store-api-55b44fb4b5-wf52z   1/1     Running   0          13s   192.168.29.203   fiap-store-worker
store-api-55b44fb4b5-4xmx5   1/1     Running   0          13s   192.168.13.15    fiap-store-worker2
store-api-55b44fb4b5-55sr6   1/1     Running   0          26s   192.168.13.13    fiap-store-worker2
store-api-55b44fb4b5-m767m   1/1     Running   0          26s   192.168.13.14    fiap-store-worker2
```

Cenário (B): topologySpreadConstraints

```
$ kubectl -n store-dev get pods -o wide --sort-by=.spec.nodeName   # 02-topologyspread.yaml (6
replicas)
NAME                         READY   STATUS    RESTARTS   AGE   IP               NODE
store-api-5979b4bb9-hnnth   1/1     Running   0          14s   192.168.163.68   fiap-store-control-
plane
store-api-5979b4bb9-xfgkq   1/1     Running   0          14s   192.168.163.69   fiap-store-control-
plane
store-api-5979b4bb9-s8gfm   1/1     Running   0          14s   192.168.29.205   fiap-store-worker
store-api-5979b4bb9-xzngg   1/1     Running   0          14s   192.168.29.206   fiap-store-worker
store-api-5979b4bb9-hjpvk   1/1     Running   0          14s   192.168.13.17    fiap-store-worker2
store-api-5979b4bb9-r76n9   1/1     Running   0          14s   192.168.13.16    fiap-store-worker2

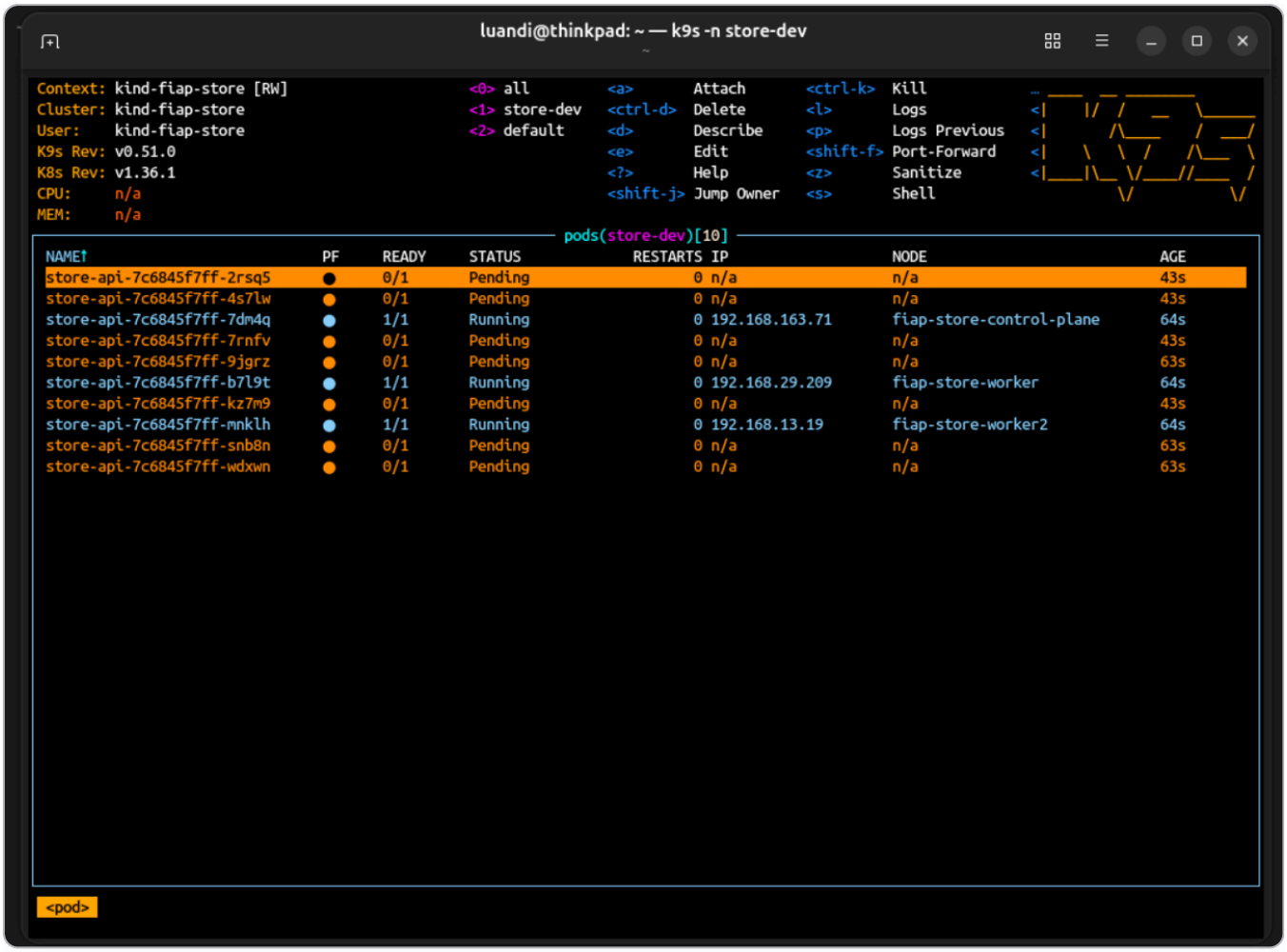
$ kubectl -n store-dev get pods -o wide --sort-by=.spec.nodeName   # 02-topologyspread.yaml (10
replicas)
NAME                         READY   STATUS    RESTARTS   AGE     IP               NODE
store-api-5979b4bb9-hnnth   1/1     Running   0          4m17s   192.168.163.68   fiap-store-control-
plane
store-api-5979b4bb9-ldk8s   1/1     Running   0          12s     192.168.163.70   fiap-store-control-
plane
store-api-5979b4bb9-xfgkq   1/1     Running   0          4m17s   192.168.163.69   fiap-store-control-
plane
store-api-5979b4bb9-9nk6q   1/1     Running   0          12s     192.168.29.207   fiap-store-worker
store-api-5979b4bb9-s8gfm   1/1     Running   0          4m17s   192.168.29.205   fiap-store-worker
store-api-5979b4bb9-x94rm   1/1     Running   0          12s     192.168.29.208   fiap-store-worker
store-api-5979b4bb9-xzngg   1/1     Running   0          4m17s   192.168.29.206   fiap-store-worker
store-api-5979b4bb9-cqvbw   1/1     Running   0          12s     192.168.13.18    fiap-store-worker2
store-api-5979b4bb9-hjpvk   1/1     Running   0          4m17s   192.168.13.17    fiap-store-worker2
store-api-5979b4bb9-r76n9   1/1     Running   0          4m17s   192.168.13.16    fiap-store-worker2
```


luandi@thinkpad: ~ — k9s -n store-dev
Context: kind-fiap-store [RW]
Cluster: kind-fiap-store
User: kind-fiap-store
K9s Rev: v0.51.0
K8s Rev: v1.36.1
CPU: n/a
MEM: n/a
<0> all
<1> store-dev
<2> default
<a> Attach
<ctrl-d> Delete
<d> Describe
<e> Edit
<?> Help
<shift-j> Jump Owner
<ctrl-k> Kill
<l> Logs
<p> Logs Previous
<shift-f> Port-Forward
<z> Sanitize
<s> Shell
pods(store-dev)[6]
NAME↑ PF READY STATUS RESTARTS IP NODE AGE
store-api-5979b4bb9-hjpvk 1/1 Running 0 192.168.13.17 fiap-store-worker2 3m50s
store-api-5979b4bb9-hnnth 1/1 Running 0 192.168.163.68 fiap-store-control-plane 3m50s
store-api-5979b4bb9-r76n9 1/1 Running 0 192.168.13.16 fiap-store-worker2 3m50s
store-api-5979b4bb9-s8gfm 1/1 Running 0 192.168.29.205 fiap-store-worker 3m50s
store-api-5979b4bb9-xfgkq 1/1 Running 0 192.168.163.69 fiap-store-control-plane 3m50s
store-api-5979b4bb9-xzngg 1/1 Running 0 192.168.29.206 fiap-store-worker 3m50s
<pod>

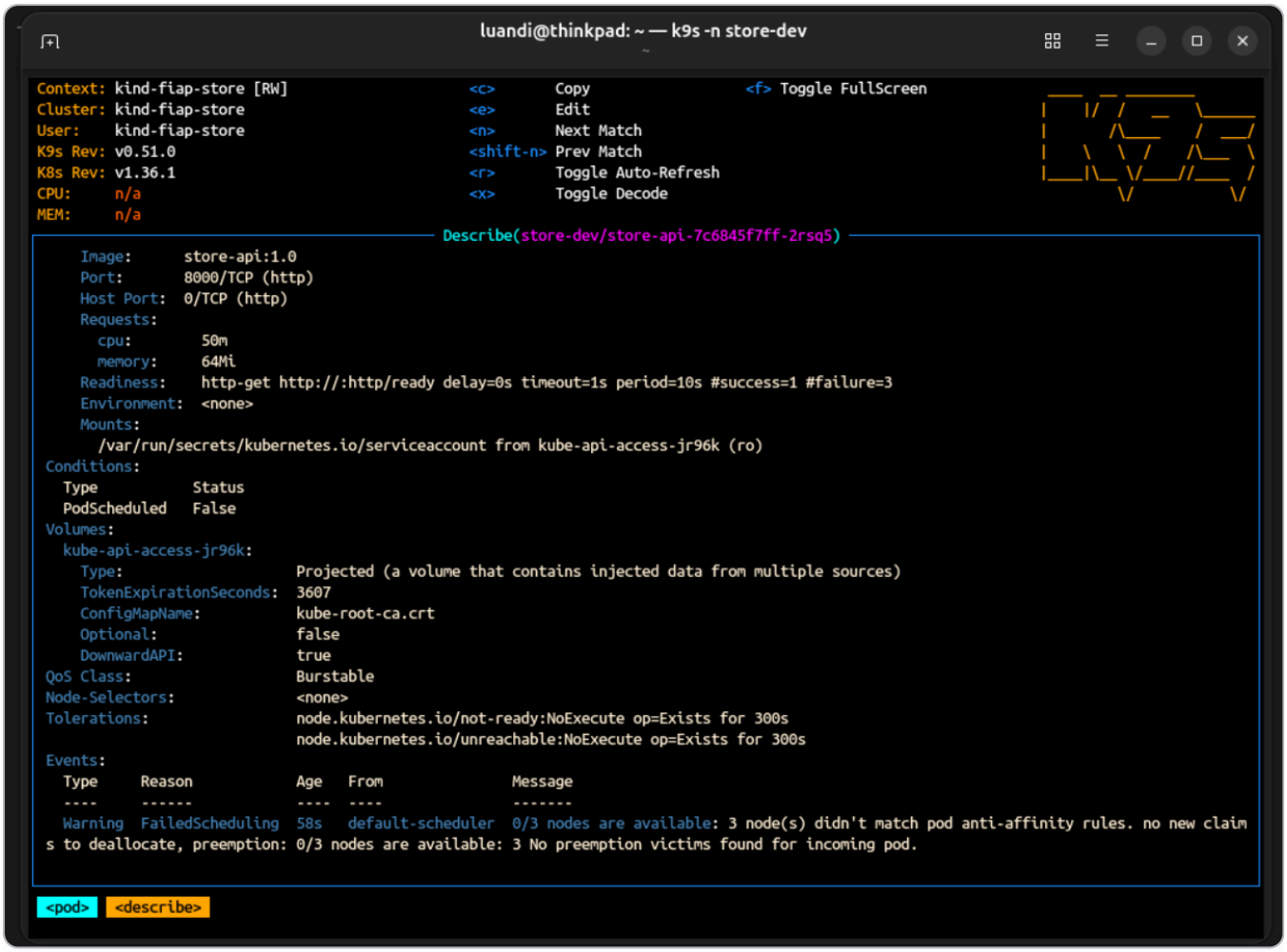
Cenário (C): podAntiAffinity required

```
$ kubectl -n store-dev get pods -o wide --sort-by=.spec.nodeName   # 03-required.yaml (6 replicas)
NAME                         READY   STATUS    RESTARTS   AGE   IP               NODE
store-api-7c6845f7ff-wdxwn   0/1     Pending   0          20s   <none>           <none>
store-api-7c6845f7ff-snb8n   0/1     Pending   0          20s   <none>           <none>
store-api-7c6845f7ff-9jgrz   0/1     Pending   0          20s   <none>           <none>
store-api-7c6845f7ff-7dm4q   1/1     Running   0          21s   192.168.163.71   fiap-store-control-
plane
store-api-7c6845f7ff-b7l9t   1/1     Running   0          21s   192.168.29.209   fiap-store-worker
store-api-7c6845f7ff-mnklh   1/1     Running   0          21s   192.168.13.19    fiap-store-worker2

$ kubectl -n store-dev get pods -o wide --sort-by=.spec.nodeName   # 03-required.yaml (10 replicas)
NAME                         READY   STATUS    RESTARTS   AGE   IP               NODE
store-api-7c6845f7ff-wdxwn   0/1     Pending   0          40s   <none>           <none>
store-api-7c6845f7ff-snb8n   0/1     Pending   0          40s   <none>           <none>
store-api-7c6845f7ff-kz7m9   0/1     Pending   0          20s   <none>           <none>
store-api-7c6845f7ff-9jgrz   0/1     Pending   0          40s   <none>           <none>
store-api-7c6845f7ff-7rnfv   0/1     Pending   0          20s   <none>           <none>
store-api-7c6845f7ff-4s7lw   0/1     Pending   0          20s   <none>           <none>
store-api-7c6845f7ff-2rsq5   0/1     Pending   0          20s   <none>           <none>
store-api-7c6845f7ff-7dm4q   1/1     Running   0          41s   192.168.163.71   fiap-store-control-
plane
store-api-7c6845f7ff-b7l9t   1/1     Running   0          41s   192.168.29.209   fiap-store-worker
store-api-7c6845f7ff-mnklh   1/1     Running   0          41s   192.168.13.19    fiap-store-worker2
```




```
$ kubectl -n store-dev describe pod store-api-7c6845f7ff-2rsq5
...
Events:
  Type     Reason            Age   From               Message
  ----     ------            ----  ----               -------
  Warning  FailedScheduling  20s   default-scheduler  0/3 nodes are available: 3 node(s) didn't match pod anti-affinity rules. no new claims to deallocate, preemption: 0/3 nodes are available: 3 No preemption victims found for incoming pod.
```



A mensagem `didn't match pod anti-affinity rules` confirma que os 3 nós já têm um Pod com `app=store-api` e a regra obrigatória impede o segundo.